DOROTEA MONACO

Software Engineer, AI Enthusiast

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PROFESSIONAL PROFILE

Master's student in Software Engineering at Politecnico di Torino with interests in AI, cloud technologies, and software development. Experienced in developing assistive solutions through volunteer work with Associazione Italiana Sclerosi Multipla, focusing on creating practical and human-centered technologies. Eager to apply technical skills to projects that combine innovation and social impact.

EDUCATION

09/2025
Turin

MSc Computer Engineering - Software
Politecnico di Torino
- Advanced training in software system design, architecture, and development
- Coursework in machine learning, computer vision, large language models, and data science
- Experience with system programming, distributed systems, databases, and network technologies
- Focus on secure, scalable, and intelligent software systems

09/2021 – 03/2025
Turin, 98

BSc Computer Engineering
Politecnico di Torino
- Algorithms and Data Structures
- Programming Techniques
- Computer Architecture
- Databases
- Operating Systems
- Computer Networks

Messina, 100

High School Diploma
Liceo Scientifico Scienze Applicate - G. Seguenza

WORK EXPERIENCE

03/2024 – 07/2024
Turin

Comune di Torino, Circoscrizione 8
Internship
- Contributed to the digitization of the municipal archive •Supported website maintenance
- Provided digital support
- Created digital content to improve communication and accessibility of public information

2023
Turin

Associazione Italiana Sclerosi Multipla, Sezione di Torino
Volunteer, Member
- During my volunteering experience, I was actively involved in the organization of informative events, fundraising activities, and awareness campaigns in companies and schools
- My role included coordinating logistics, supporting communication with participants, and helping promote the organization's mission to different audiences

SKILLS

Soft Skills

- Teamwork
- Problem Solving
- Organization & Time Management
- Resilience & Adaptability
- Continuous Learning

Technical Skills

- Programming Languages: Python, Java, C, Rust, JavaScript, TypeScript, SQL, Assembly
- Web Development: HTML, CSS, WebSockets
- Databases: PostgreSQL, MySQL, Redis
- Software Engineering & Systems: Software Architecture, System Programming, Formal Languages & Compilers, Information Systems
- Machine Learning & AI: Machine Learning, Computer Vision, Large Language Models for Software Engineering
- Networking & Security
- Tools & Technologies: Git, GitHub, Linux, REST APIs

LANGUAGES

English
IELTS 6.0

Italian
Native

PROJECTS

12/2025 – 02/2026

GAN for Data Augmentation and Domain Adaptation
Technologies: Python, PyTorch/TensorFlow, Deep Learning, GANs, Data Augmentation
Developed a deep learning pipeline using Generative Adversarial Networks (GANs) to generate synthetic data and improve model performance in scenarios with limited or imbalanced datasets. Implemented domain adaptation techniques to reduce distribution differences between source and target datasets, enabling better generalization of machine learning models. GAN-based augmentation is commonly used to create additional training samples and improve robustness when data is scarce.

12/2025 – 01/2026

Architectures for Code Development with LLMs
Technologies: Python, Large Language Models (LLMs), AI-assisted software development
Designed and evaluated software architectures that integrate Large Language Models (LLMs) into the software development workflow. Explored approaches for leveraging LLMs to assist in code generation, analysis, and development support, focusing on structuring systems that effectively combine AI components with traditional software engineering pipelines. LLMbased tools can significantly enhance developer productivity by generating or completing code and supporting programming tasks.

10/2025 – 01/2026

Participium
Technologies: React, TypeScript, Node.js, Express, PostgreSQL, Docker, RESTful APIs, HTML5, CSS3, Git, Jest (for testing)
Developed a full-stack web application enabling citizens to report urban issues (e.g., potholes, broken streetlights, waste) directly to municipal offices via an intuitive map-based interface. The platform features role-based access control, interactive geolocation, real-time status tracking, secure authentication, responsive UI, and automated tests. The project was executed using Scrum methodology, demonstrating skills in full-stack development, API design, containerized deployment, and collaborative software engineering in an academic team project.

10/2025

Machine Learning Projects

Technologies: Python, Scikit-learn, Pandas, NumPy, Jupyter Notebook, Data Analysis, Machine Learning

Collection of machine learning experiments and implementations exploring data preprocessing, model training, and evaluation across different predictive tasks. The projects apply supervised learning techniques and common ML workflows—including feature engineering, model comparison, and performance evaluation—to gain practical experience in building and analyzing machine learning models. Machine learning workflows typically involve steps such as data preparation, model training, and validation to improve predictive performance and generalization.

07/2025 – 09/2025

Ruggine

Technologies: Rust, WebSocket, Tokio (async runtime), Iced (desktop GUI), SQLite/PostgreSQL, Redis, Docker, Git, AES-256-GCM encryption.

Developed a modern, secure, and scalable real-time chat application built entirely in Rust with a cross-platform desktop client. The system supports end-to-end encrypted messaging, WebSocket-based real-time communication, private and group chats, persistent message history, online presence tracking, and efficient asynchronous server architecture.

OPEN SOURCE

02/2026

LLM4S

Contributed multiple improvements and features to the llm4s open-source project, including:

- Fixed broken and missing documentation links to improve usability and onboarding.
- Added unit tests for ChunkerFactory, DocumentChunk, and SimpleChunk modules to ensure code reliability.
- Implemented safe embedding validation in embedBatch to enhance robustness and prevent runtime errors.
- Developed the AdaptiveWindowing pruning strategy with intelligent context window autotuning, including detailed documentation and production-ready examples.

These contributions demonstrate skills in Scala development, testing, documentation, realtime data processing, and collaborative open-source workflow using GitHub Pull Requests

02/2026

MoFA

Contributed multiple enhancements and fixes through merged pull requests, including:

- Improved build infrastructure and CI support to streamline development workflows.
- Refined command utilities and simplified configuration parsing for better developer experience.
- Fixed critical bugs, strengthened type safety, and resolved edge-case panics to improve stability.
- Enhanced feature support in core modules, expanding the agent runtime's capability set.
- Added comprehensive tests and validations to increase code quality and regression safeguards.
- Upgraded documentation and examples to help new contributors understand core abstractions.

These contributions demonstrate expertise in systems programming (Rust), open-source collaboration, distributed development workflows, maintainability improvements, and quality-driven software engineering through community review and iterative refinement.